CE60258 AI/ML for Sustainable Infrastructure Systems — One-Month Integrated Big Problem

Integrated study of a 33 m post-tensioned PSC I-girder bridge: design, baseline, inspection, monitoring, AI/ML condition assessment and maintenance

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Submission	October 2026

Summary

A 33 m post-tensioned PSC I-girder bridge (3 lanes, footpaths, 6 girders at 2.6 m, 1.80 m + 0.22 m deck, M45/M35) was designed to IRC:6 and IRC:112 with two-stage post-tensioning (4 × 11T15; P_e = 6538 kN, total loss 23.9 %; all stress, ULS, shear and deflection checks satisfied). A digital twin of the as-built bridge gives the healthy baseline (f_1 = 3.44 Hz bending, 3.51 Hz torsion; -0.17 %/°C), the inspection, NDT and monitoring schemes, and 2380 synthetic campaigns for six deterioration mechanisms. On these data, vibration features alone classify the mechanism with macro-F1 0.41; fusing visual inspection, NDT, monitoring and vibration gives 0.91. Prestress loss is estimated from levelling and embedded gauges with an error of 3.6 % of P_e. A rule-based maintenance scheme turns the fused evidence into auditable decisions; over a simulated 30-year life the proactive owner keeps every mechanism below severity 0.6 at a total of 32 relative units, against 51 (including the unrepaired damage) for a reactive owner, who never sees the grout-void prestress loss. All data are synthetic: the ML figures are digital-twin performance.

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1 Introduction and bridge configuration

This report follows one bridge through its life: a simply supported, post-tensioned PSC I-girder span of **L = 33 m** (span chosen from 30 / 33 / 36 m), with three traffic lanes, footpaths and crash barriers on both sides. The chain of the problem statement is followed literally: design → prestressing → healthy baseline → visual inspection → NDT → structural and vibration monitoring → synthetic data → features → AI/ML → maintenance decision. One digital twin of the bridge (a grillage with mass, bearings and damage hooks) produces all the synthetic measurements, so every later chapter uses the same structure designed in Chapters 2–3.

The span is treated as span S1 (abutment A1 – pier P1) of a viaduct of identical simply supported spans, so that the inspection scheme can include a pier cap and a pier; piers, abutments and foundations are not designed here.

1.1 Cross-section

Strip	Width (m)	Justification
Railing + kerb	2 × 0.25	pedestrian railing on a kerb
Footpath	2 × 1.50	IRC:5 minimum clear footpath width
Crash barrier	2 × 0.45	rigid New-Jersey barrier separating footpath and carriageway
Carriageway	11.00	IRC:6 Table 6: 9.6 ≤ 11.0 < 13.1 m gives exactly three design lanes
Overall deck	15.40	

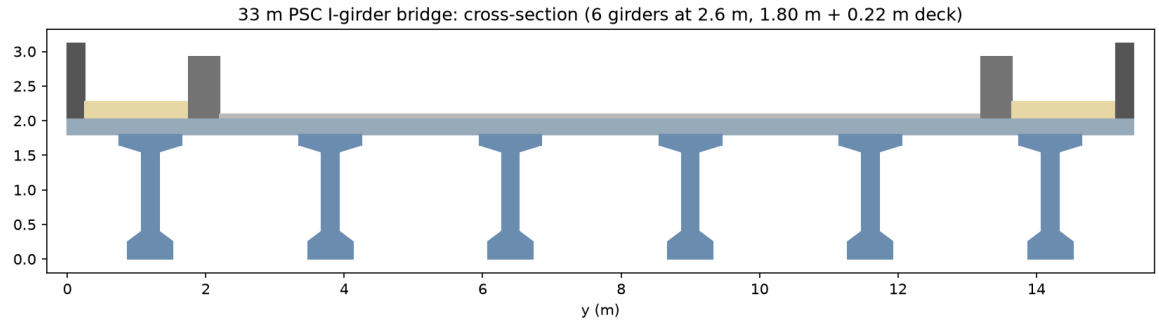


Figure 1. Adopted cross-section: six PSC I-girders at 2.6 m, 1.80 m precast + 0.22 m in-situ deck.

Girders. Six girders at 2.6 m with 1.20 m cantilevers. Compared with five girders at about 3.1 m, six lighter girders (precast area 0.710 m², about 61 t each) are easier to erect and allow a shallower section. Depth 1.80 m precast + 0.22 m deck = 2.02 m (L/D = 16.3). Web 0.25 m (one 12-strand duct + 75 mm cover each side), bottom flange 0.65 × 0.25 m to house the cables, end blocks widened to 0.65 m over 1.5 m for the anchorages. **Concrete** M45 girders and M35 deck (IRC:112 minimum grades for severe exposure are M40 PSC and M30 RCC); E_{cm} = 34.3 / 32.3 GPa (IRC:112 Table 6.5). **Cross girders** at both ends (0.50 × 1.50 m, also the jacking beams for bearing replacement) and one at mid-span (0.30 × 1.40 m).

Section properties (composite in girder concrete, modular ratio 0.942; the full deck width is effective by IRC:112 cl. 7.6.1.2).

Section	A (m²)	y_b (m)	I (m⁴)	Z_b (m³)	Z_t (m³)
precast mid	0.7100	0.912	0.2761	0.3029	0.3108
precast end block	1.2200	0.933	0.3466	0.3716	0.3997
composite mid, interior	1.2486	1.342	0.5835	0.4347	0.8610
composite mid, outer	1.2279	1.333	0.5767	0.4327	0.8390

2 Loads and structural analysis

Loads (IRC:6-2017). Self-weight (25 kN/m³); crash barriers 8.5 kN/m, railings 2.0 kN/m, footpath finish 3.0 kPa, wearing coat 2.0 kPa (65 mm bituminous plus one overlay). Footpath live load by cl. 206.1 for L = 33 m: 2.80 kPa. Vehicles: Class A trains, Class 70R wheeled and tracked, combined per Table 6A for three lanes (one 70R + one Class A, or three Class A with the 10 % reduction of cl. 205), with the IRC kerb and vehicle-to-vehicle clearances; impact 4.5/(6 + L) = 0.115 for Class A and wheeled 70R, 10 % for tracked 70R.

Model. Stage 1 (girder + wet slab + cross girders) on the precast girder alone as a simply supported beam; composite loads on a grillage (6 longitudinal composite members, transverse slab strips at 0.75 m, cross girders as T-beams, Hambly's torsion split). Influence surfaces are obtained with one solve per response (Maxwell–Betti), and the vehicles are placed by an exhaustive longitudinal search plus dynamic programming across the carriageway.

Checks of the tools. The grillage reproduces PL/4, PL³/48EI and wL²/8 for a single line and static equilibrium for the full deck; the vehicle search reproduces its own optimum by a direct load-vector solve. For the governing outer-girder case (70R wheeled + Class A, eccentricity -1.57 m) the grillage gives a share of 0.251 to G1 against 0.253 by Courbon's method (L/B = 2.14, stiff mid-span cross girder): the two methods agree.

Characteristic mid-span moments (kNm). ULS = 1.35 (DL + SIDL) + 1.75 WC + 1.5 (LL + FP), IRC:6 Annex B.

Girder	DL stage 1	SIDL	Wearing coat	Footpath LL	Vehicle LL (incl. impact)	SLS rare	ULS
G1	4,403	653	503	261	2,735	8,655	**12,199**
G2	4,580	739	486	244	2,424	8,569	**12,032**
G3	4,580	650	509	174	2,116	8,130	**11,385**

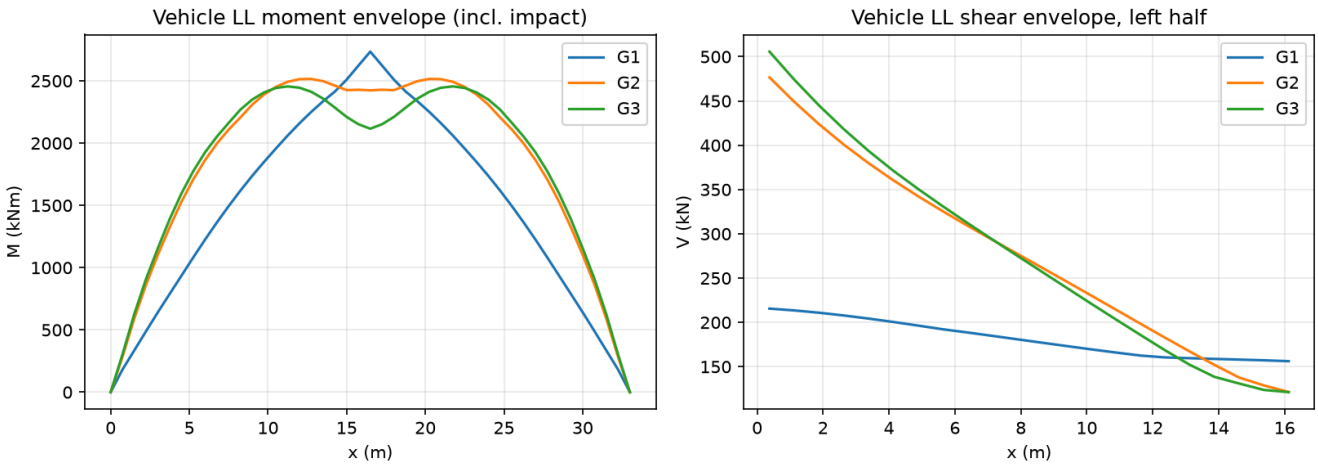


Figure 2. Vehicle live-load envelopes. The mid-span cross girder pushes load outwards at mid-span, so G2/G3 peak near 0.37 L while G1 peaks at mid-span.

The outer girder G1 governs flexure (live-load deflection 13.1 mm < L/800 = 41.2 mm); interior girders govern shear (vehicle shear 506 kN at the support element). All six girders are given the same design.

3 Prestressing system

Two-stage stressing. With all strands stressed on the bare girder the transfer compression at the soffit exceeds 0.48 $f_{ck}(t)$ for concrete of about 35 MPa at 14 days. Two cables are therefore stressed at a girder age of 14 days and two after the deck has hardened (day 75), when they act on the deeper composite section. The strand count was chosen by a sweep: 3 × 12T15 and 4 × 10T15 crack under the frequent combination (with the characteristic prestress 0.9 P_m), 4 × 12T15 exceeds 0.36 f_{ck} under quasi-permanent load near the supports; **4 × 11T15 (44 strands, 15.2 mm, f_{pk} 1860 MPa)** satisfies every check.

Cable layout. Parabolic profiles with a straight run; lateral S-curves inside the bottom flange.

Cable	Strands	Stage	Anchor height (m)	Mid-span height (m)	Mid-span lateral (m)
C1	11T15	1	0.40	0.125	-0.19
C2	11T15	1	0.75	0.125	+0.19
C3	11T15	2	1.10	0.125	+0.00
C4	11T15	2	1.45	0.315	+0.00

The geometry was checked along the whole girder: minimum duct centre distance 190 mm (≥ 2 × OD = 190 mm), minimum cover 78 mm (≥ 75 mm), anchorage spacing 0.35 m with 0.35 m top edge distance, minimum plan radius 5.5 m (to be confirmed with the PT supplier).

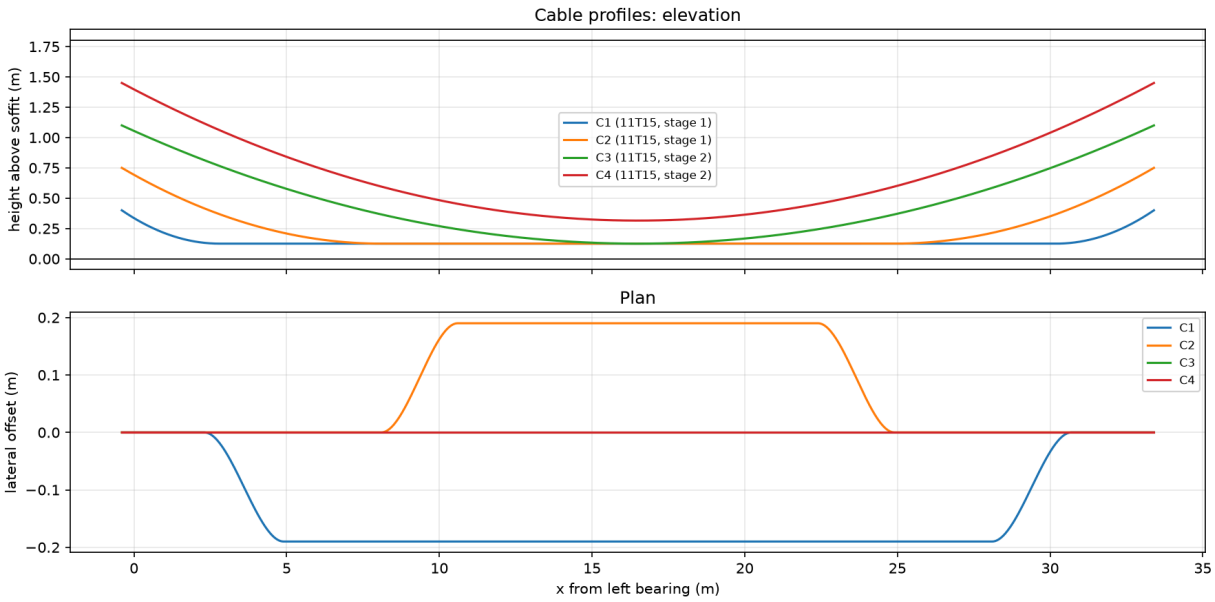


Figure 3. Cable profiles in elevation and plan.

Prestress losses: P_j = 8593 kN, P_0 = 8008 kN, P_e = 6538 kN, total loss 23.9 % (IRC:112 cl. 7.9 and Annex A2).

Stage of loss (mid-span, G1)	Stress (MPa)	Note
Jacking stress (0.75 f_{pk})	1,395	–
After friction (μ = 0.17, k = 0.002/m) and 6 mm wedge slip, stage-1 cables	1,286	both ends stressed
Elastic shortening, stage-1 cables stressed in sequence	12.2	(n−1)/2n rule
Elastic shortening of stage-1 cables when stage 2 is stressed	70.9	cost of two-stage stressing
Creep + shrinkage + relaxation, stage 1	213.0	φ = 1.89
Creep + shrinkage + relaxation, stage 2	193.3	φ = 1.37
Final stress stage 1 / stage 2	990 / 1133	

Stresses. Construction stages (transfer, deck casting, stage-2 stressing) and service at opening and after 100 years were checked with the characteristic prestress r_{sup} = 1.1 / r_{inf} = 0.9 P_m, temperature gradients (IRC:6 cl. 215.3; mid-span soffit +1.76 / -2.46 MPa) and differential shrinkage (deck 334 με, girder after deck casting 229 με, age-adjusted modulus). Severe exposure, bonded PT: decompression under the quasi-permanent combination and $w_k \le 0.2$ mm under the frequent combination (the section stays uncracked, $w_k = 0$).

Governing stress checks over all stations of girders G1–G3 (compression positive).

Stage	Fibre	Check	Worst value (MPa)	Limit	Result	Where
transfer	girder_bottom	compression <= 0.48 fck(t)	13.13	19.00	OK	G1, x = 3.00
transfer	girder_bottom	tension >= -fctm(t)	6.62	-2.95	OK	G1, x = 0.00
transfer	girder_top	compression <= 0.48 fck(t)	3.68	19.00	OK	G1, x = 16.50
transfer	girder_top	tension >= -fctm(t)	-1.44	-2.95	OK	G1, x = 1.50
deck_cast	girder_bottom	compression <= 0.48 fck(t)	9.75	21.60	OK	G1, x = 2.25
deck_cast	girder_bottom	tension >= -fctm(t)	-1.93	-3.45	OK	G3, x = 16.50
deck_cast	girder_top	compression <= 0.48 fck(t)	11.04	21.60	OK	G3, x = 16.50
deck_cast	girder_top	tension >= -fctm(t)	-0.85	-3.45	OK	G2, x = 0.00
stage2	girder_bottom	compression <= 0.48 fck(t)	16.85	21.60	OK	G1, x = 5.25
stage2	girder_bottom	tension >= -fctm(t)	8.02	-3.61	OK	G1, x = 0.00
stage2	girder_top	compression <= 0.48 fck(t)	10.78	21.60	OK	G3, x = 16.50
stage2	girder_top	tension >= -fctm(t)	1.29	-3.61	OK	G2, x = 0.75
open/rare	girder_bottom	compression <= 0.48 fck	14.80	21.60	OK	G1, x = 2.25
open/rare	girder_bottom	[legacy] no tension under rare	-1.76	0.00	not required (IRC:112)	G1, x = 16.50
open/rare	girder_top	compression <= 0.48 fck	14.50	21.60	OK	G2, x = 16.50
open/rare	deck_top	compression <= 0.48 fck	5.67	16.80	OK	G1, x = 16.50
open/frequent	girder_bottom	uncracked: tension >= -fctm	0.12	-3.28	OK	G2, x = 16.50
open/frequent	girder_top	uncracked: tension >= -fctm	0.71	-3.28	OK	G2, x = 0.00
open/frequent	deck_top	uncracked: tension >= -fctm	0.31	-2.77	OK	G3, x = 16.50
open/QP	girder_bottom	compression <= 0.36 fck	16.07	16.20	OK	G3, x = 2.25
open/QP	girder_bottom	decompression at soffit	10.38	0.00	OK	G3, x = 2.25
open/QP	girder_top	compression <= 0.36 fck	12.31	16.20	OK	G2, x = 16.50
open/QP	deck_top	compression <= 0.36 fck	4.47	12.60	OK	G1, x = 0.00
final/rare	girder_bottom	compression <= 0.48 fck	12.66	21.60	OK	G1, x = 2.25
final/rare	girder_bottom	[legacy] no tension under rare	-4.32	0.00	not required (IRC:112)	G1, x = 16.50
final/rare	girder_top	compression <= 0.48 fck	15.37	21.60	OK	G2, x = 16.50
final/rare	deck_top	compression <= 0.48 fck	5.76	16.80	OK	G1, x = 16.50
final/frequent	girder_bottom	uncracked: tension >= -fctm	-2.39	-3.28	OK	G2, x = 16.50
final/frequent	girder_top	uncracked: tension >= -fctm	1.27	-3.28	OK	G2, x = 0.00
final/frequent	deck_top	uncracked: tension >= -fctm	0.23	-2.77	OK	G2, x = 0.00
final/QP	girder_bottom	compression <= 0.36 fck	13.92	16.20	OK	G3, x = 2.25
final/QP	girder_bottom	decompression at soffit	8.55	0.00	OK	G3, x = 2.25
final/QP	girder_top	compression <= 0.36 fck	13.18	16.20	OK	G2, x = 16.50
final/QP	deck_top	compression <= 0.36 fck	4.02	12.60	OK	G1, x = 0.00

ULS. Strain-compatibility fibre analysis (parabola–rectangle concrete, bilinear strand with its effective prestrain, untensioned bars ignored): M_{Ed}/M_{Rd} = 0.81 (G1, 12199 / 15031 kNm), neutral axis 291 mm below the deck top, all tendons yielded. A hand check with the parabola–rectangle block agrees within 1 %. **Shear** (IRC:112 cl. 10.3, interior girder): V_{Ed} = 1690 kN at the support, of which the cable inclination carries 747 kN;

links were designed by the truss model alone (concrete term ignored): A_{sw}/s = 693 mm²/m near the support (e.g. 2L-12 @ 300) reducing to the minimum 241 mm²/m at mid-span; $V_{Rd,max}$ is not governing. Deck–girder interface steel up to 1831 mm²/m near the supports. **Deflection**: camber 14.6 mm at transfer, 3.6 mm upwards in the long term; live load 15.3 mm (beam line) against L/800 = 41.2 mm.

4 Healthy baseline

The as-built twin uses the dynamic modulus $E_d = (E_c + 19)/1.25$ GPa (BS 8110-2; 1.24 E_{cm}), the superimposed masses (27.1 t/m in total, equal to the dead loads within 1 %) and laminated elastomeric bearings 400 × 400 mm (shape factor 8.3, k_v = 1149 MN/m, k_h = 3333 kN/m) as springs. Damping 1.5 % in every mode.

Natural frequencies of the healthy bridge at 30 °C. Hand check: simply supported beam f_1 = 3.38 Hz.

Mode	f (Hz)	Shape
1	3.442	bending 1
2	3.510	torsion 1
3	7.958	transverse 1
4	12.967	torsion 2
5	13.018	bending 2
6	14.088	transverse 2

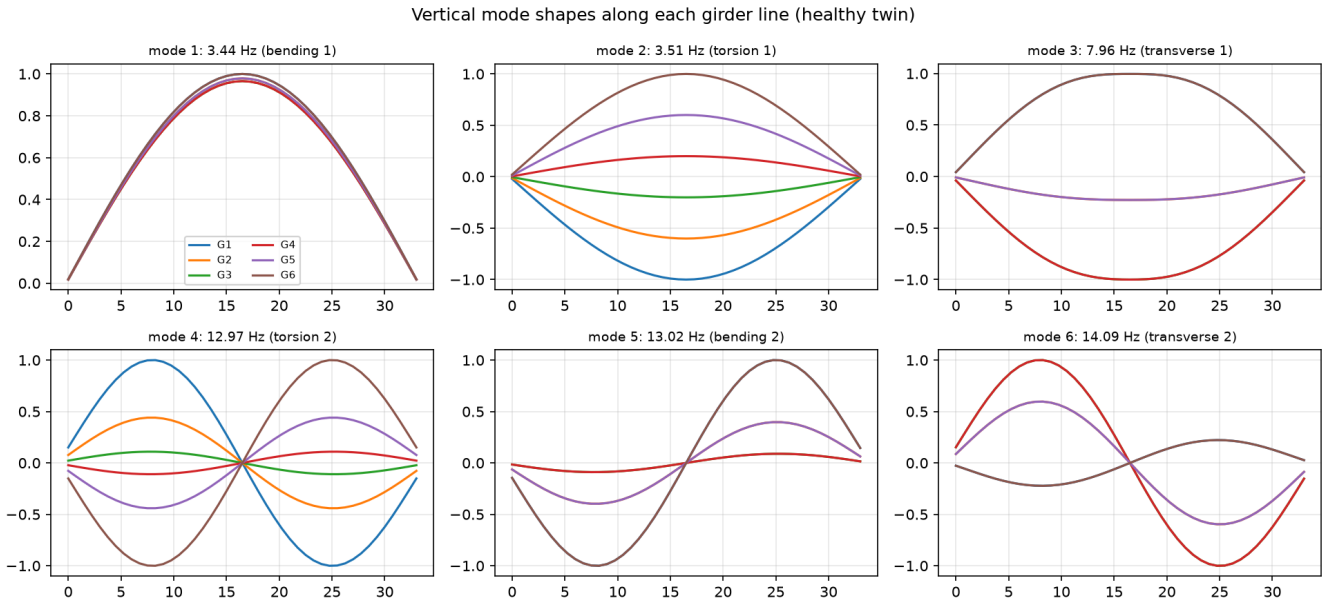


Figure 4. Vertical mode shapes along each girder line.

Vibration measurement. Eight vertical accelerometers (L/4, L/2, 3L/4 of both outer girders and mid-span of G3 and G4; the outer pairs separate bending from torsion), 100 Hz sampling (Nyquist 50 Hz ≫ the 25 Hz band of interest), 10-minute records (about 2000 cycles of f_1), excitation by traffic and ambient wind / micro-tremor. Processing: detrend → 0.5–30 Hz band-pass → Welch PSD → peak picking; frequency domain decomposition (FDD) of the cross-spectral matrix for the mode shapes, compared with the model by MAC.

Identification of the healthy bridge from simulated 10-minute records (traffic + ambient + sensor noise).

Mode	Model (Hz)	Identified from 3 simulated records (Hz)	MAC
bending 1	3.442	3.458 ± 0.006	0.93
torsion 1	3.510	3.510 ± 0.022	0.89
transverse 1	7.958	7.944 ± 0.028	1.00

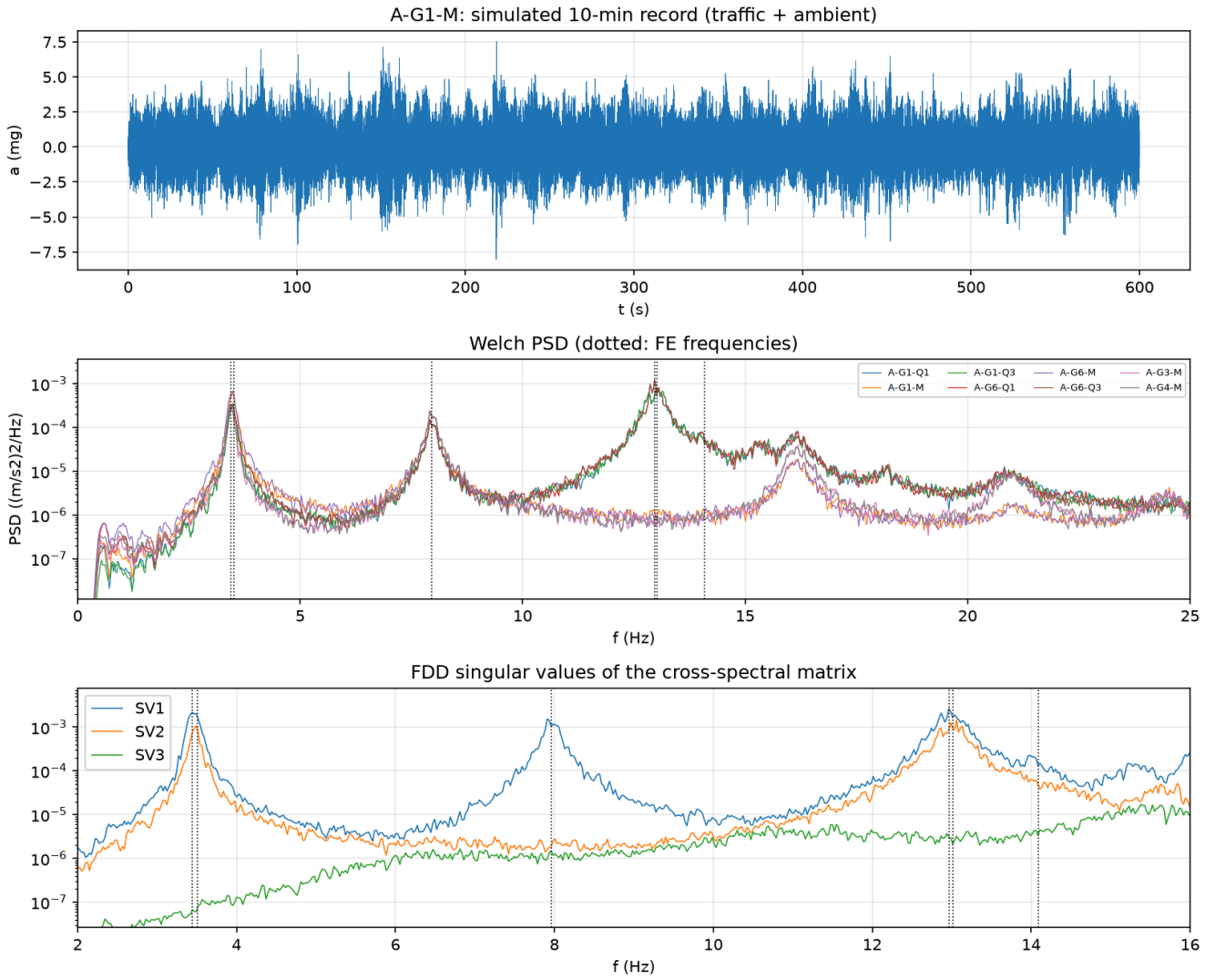


Figure 5. Simulated acceleration record, Welch PSD of all channels and FDD singular values.

The first bending and first torsion modes are only 2 % apart, so the identified f_1 carries a small bias and the 13 Hz pair cannot be separated at all; **the baseline for damage detection must therefore be the identified values, not the model**. Sensitivities: temperature -0.170 %/°C on f_1 (a 35 °C seasonal swing moves f_1 by about 6 %); fully composite barriers +0.7 %; seized bearings +8.5 % – a seized bearing makes the bridge look *stiffer*.

Static baseline. Load test with two 400 kN trucks (IRC:SP:51 approach), mid-span values G1–G6:

Test	Deflection (mm)	Soffit strain (με)
LT-1 eccentric (lanes 1-2)	8.37 / 7.51 / 6.25 / 4.51 / 2.56 / 0.60	119 / 102 / 84 / 64 / 37 / 6
LT-2 central	4.01 / 5.08 / 5.81 / 5.81 / 5.08 / 4.01	56 / 72 / 78 / 78 / 72 / 56

Long-term reference curves (G3 mid-span, creep superposition of every stress increment and the loss history): embedded-gauge strain / mid-span deflection 1 y: 807 με / -3.9 mm, 10 y: 926 με / -1.0 mm, 30 y: 939 με / -0.6 mm, 100 y: 944 με / -0.5 mm. Later readings are compared with these curves.

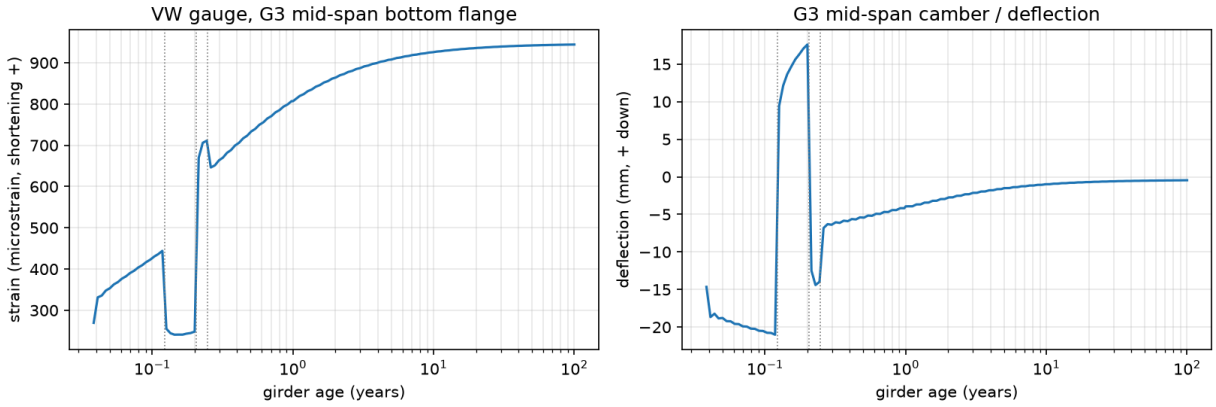


Figure 6. Reference curves of the embedded VW gauge and the mid-span camber.

4.1 Structural and vibration monitoring scheme

31 sensors / items: 8 accelerometers; 12 embedded vibrating-wire gauges with thermistors (bottom and top flange of G1, G3, G6 at L/4 and L/2, read every 15 min from casting); hydrostatic levelling at mid-span of G1, G3, G4, G6; bearing-movement transducers at the free bearings of G1 and G6 (expected 0.386 mm/°C); thermocouples through the depth; a weather station; and a load test at handing over and every 10 years.

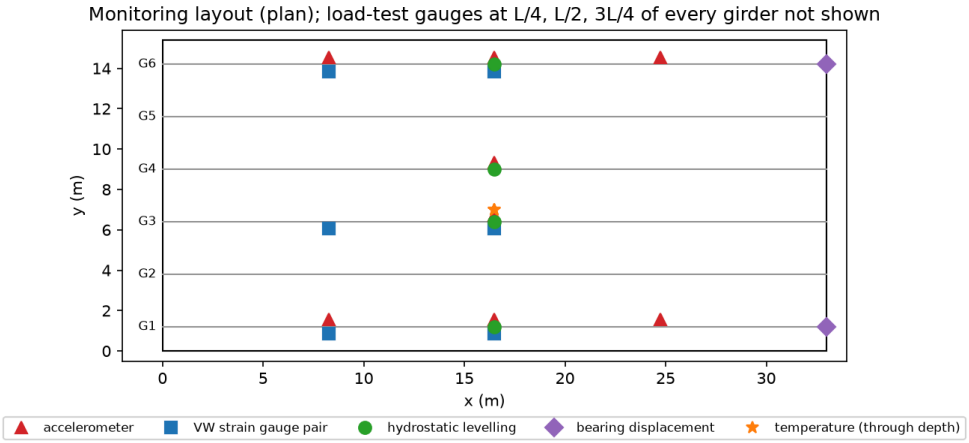


Figure 7. Monitoring layout (plan).

Healthy NDT reference values (assumed for new concrete; replaced by the survey at handing over).

NDT method	Healthy reference (mean ± sd)	Note
rebound_number	girder (M45): 41.0 ± 2.5; deck (M35): 37.0 ± 2.5	IS 516 (Part 5/Sec 4):2020; horizontal impact on the web, vertical upward on the soffit
upv_km_s	girder (M45): 4.6 ± 0.12; deck (M35): 4.45 ± 0.12	IS 516 (Part 5/Sec 1):2018, direct transmission through the web where possible
cover_mm	girder links (spec 50): 55.0 ± 4.0; ducts (spec 75): 80.0 ± 5.0; deck (spec 50): 54.0 ± 5.0	cover meter; tolerance +10/-0 mm
half_cell_mV_CSE	all: -120.0 ± 40.0	ASTM C876: > -200 mV: < 10 % probability of active corrosion
resistivity_kOhm_cm	girder: 35.0 ± 8.0; deck: 25.0 ± 6.0	Wenner probe; > 20 kOhm cm: low corrosion rate
carbonation_depth_mm	model: x = k sqrt(t), k = 1.5 mm/sqrt(year) for M45, 2.0 for M35	phenolphthalein on cores / drilled holes
chloride_at_steel_pct_cement	all: 0.05 ± 0.02	threshold ~0.3-0.4 % by mass of cement
duct_grouting	all: fully grouted	GPR / ultrasonic tomography over the ducts, impact-echo at anchorages; voids are a construction defect to rule out at handing over

5 Visual inspection

Hierarchy **Bridge → Span → Component → Location → Observation**: bridge BR-33, span S1, 12 components, 268 coded locations (e.g. `G3/L/2/bottom flange`, `BRG-P1-G6`, `EJ-A1/carriageway`). Each observation type has measured inputs (crack width and orientation, spall area and exposed steel, bearing shear strain, …) and a rule that gives a condition state CS1 (good) to CS4 (severe). PSC girders are rated more strictly than RCC because they were designed uncracked (a 0.1 mm flexural crack is CS3; any crack along a duct is at least CS3). A component takes its worst observation; the bridge health index HI = 100 Σ w (4 − CS)/3 / Σ w weights components by structural importance, and a CS4 on girders, bearings, pier or abutment overrides the index.

Component	Weight	Locations	Observation types
PSC girders	1.0	102	crack, spalling, corrosion staining, leakage / efflorescence…
Deck slab	0.8	70	crack, spalling, corrosion staining, leakage / efflorescence…
Cross girders	0.6	15	crack, spalling, corrosion staining, leakage / efflorescence…
Bearings	0.7	12	bearing shear deformation, bearing bulging / splitting, bearing displacement / walking, bearing seized / debris…
Expansion joints	0.5	6	joint debris, joint seal torn / leaking, joint anchorage damage
Drainage	0.3	8	drain blocked, drain damaged / missing, discharge onto structure
Crash barriers	0.4	10	crack, spalling, impact damage, exposed steel
Footpaths	0.2	10	surface deterioration, crack, spalling
Railings	0.2	10	impact damage, corrosion staining, missing members
Wearing coat	0.3	15	wearing coat cracking, potholes / ravelling, rutting
Abutment A1	0.8	5	crack, spalling, leakage / efflorescence, corrosion staining…
Pier P1	0.9	5	crack, spalling, leakage / efflorescence, corrosion staining…

Inspection	Interval	Scope
superficial (patrol)	monthly	drive-over and walk-along: joints, drains, barriers, wearing coat, visible damage
routine	twice a year (pre- and post-monsoon)	all components from deck and ground with binoculars; bearings and joints close up; app records
principal (detailed)	every 3 years and at handing over	hands-on, under-bridge unit; every location of the hierarchy; crack mapping; baseline NDT grid
special	after flood, earthquake, vehicle impact, fire, overload, or a monitoring alarm	targeted: NDT, load test, structural assessment

Digital inspection app (`inspection_app/index.html`, one offline file): clickable plan of the span (underside and deck-top views) coloured by the worst condition state, defect-specific inputs with the condition state computed live (override only with a stated reason), photos, health index, recommended NDT, CSV/JSON export. Its rules are the same as the Python rules (a test compares both on more than 500 cases).

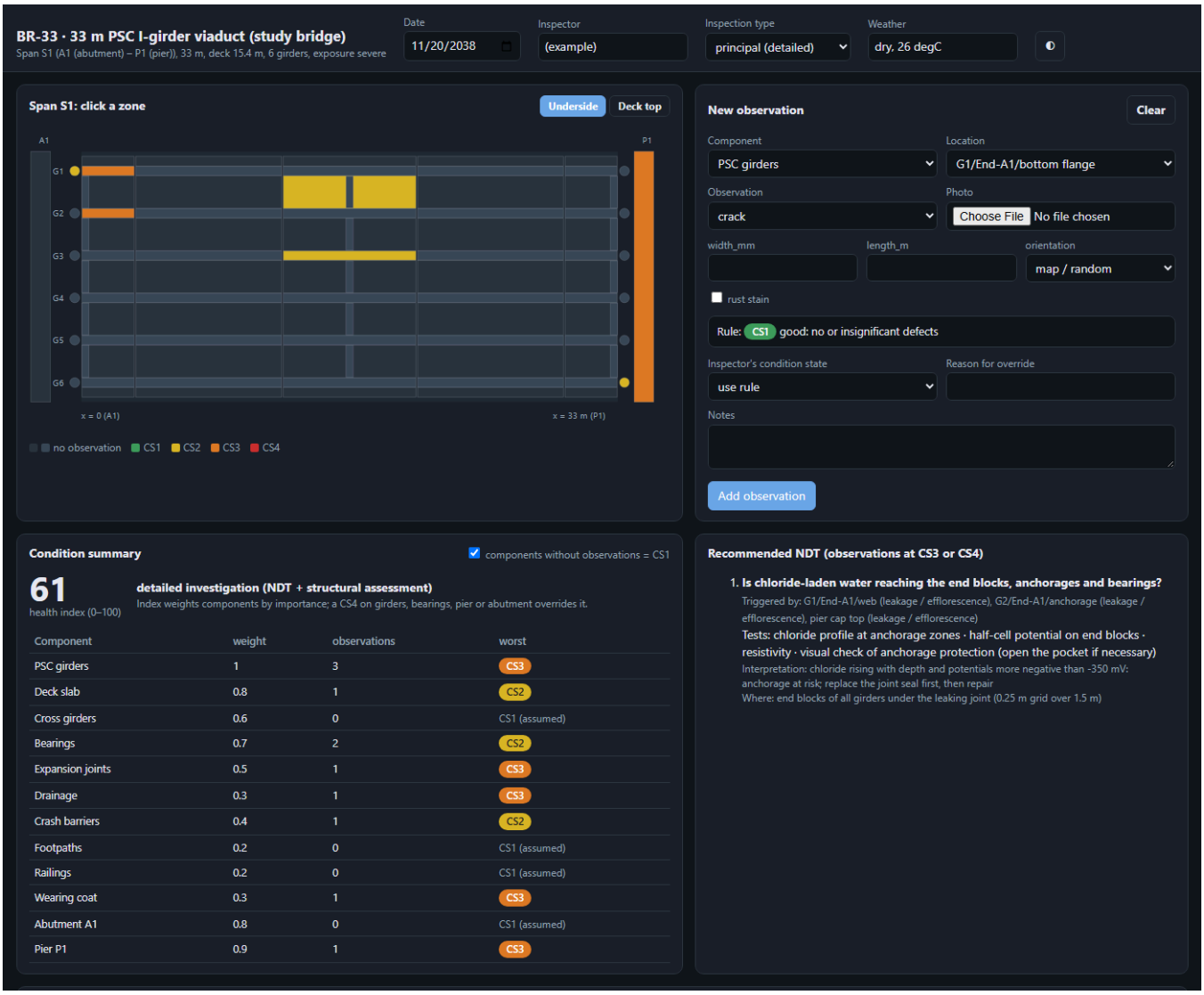


Figure 8. Inspection app with the illustrative year-12 principal inspection (leaking joint at A1 feeding efflorescence on the G1/G2 end blocks).

6 Non-destructive testing

Each test answers an engineering question raised by an observed defect (**defect → question → NDT → measurement → interpretation**). The baseline grid has 81 test points (soffit and web of every girder at five zones, deck soffit bays) where rebound, UPV, cover, half-cell potential and resistivity are measured at handing over and at principal inspections; chloride profiles are taken at the end blocks.

Observed defect	Engineering question	NDT	Interpretation
longitudinal crack / rust staining along a duct line (girder bottom flange)	Are the prestressing strands corroding? Is the duct fully grouted?	half-cell potential grid over the duct line; concrete resistivity (Wenner); cover meter; GPR / ultrasonic tomography over the duct; chloride profile from drilled dust; if indicated: endoscope through a drilled hole into the duct	potential < -350 mV CSE with resistivity < 10 kOhm cm: active corrosion likely; chloride at duct depth > 0.2 % of cement: depassivation of strand likely; tomography anomaly: grout void -> re-grouting (vacuum injection)
transverse flexural cracks at the soffit near mid-span	Has the girder lost prestress (designed uncracked)?	crack width / length mapping and tell-tales or crack gauges; UPV across the crack (crack depth); VW strain and levelling trend vs. reference curve; load test (deflection, strain, crack opening)	cracks opening under traffic and a strain / camber trend away from the reference curve indicate prestress loss; compare decompression load in the load test with design
diagonal web cracks near the supports	Is the shear capacity or the end-block reinforcement adequate?	crack mapping, width monitoring; UPV (direct, through the web); cover meter for link spacing; load test with trucks near the support	cracks > 0.2 mm or growing: detailed shear assessment with as-built links
honeycombing / surface deterioration / doubtful concrete	Is the in-situ concrete quality and strength adequate?	rebound hammer (IS 516 Part 5/Sec 4); UPV (IS 516 Part 5/Sec 1); combined rebound + UPV (SonReb) calibrated on cores (IS 516 Part 4)	UPV well below the healthy reference (e.g. < 3.75 km/s, 'doubtful' in IS 516-5-1) or rebound below reference - 2 sd: take cores; estimate strength from the calibrated curve
spalling / delamination / exposed rebar on the deck soffit or cross girders	How far does corrosion-induced delamination extend? What is causing it?	hammer sounding; half-cell potential; cover meter; carbonation depth (phenolphthalein); chloride profile	carbonation depth >= cover or chloride > 0.4 % of cement at rebar: corrosion initiated; repair extent = hollow-sounding area + 0.3 m
leakage / efflorescence at girder ends below an expansion joint	Is chloride-laden water reaching the end blocks, anchorages and bearings?	chloride profile at anchorage zones; half-cell potential on end blocks; resistivity; visual check of anchorage protection (open the pocket if necessary)	chloride rising with depth and potentials more negative than -350 mV: anchorage at risk; replace the joint seal first, then repair
bursting / splitting cracks in the anchorage zone	Is the anchorage-zone reinforcement working? Is the anchorage itself sound?	crack mapping; UPV; GPR for bursting steel; impact-echo on the anchorage pocket	cracks > 0.2 mm or voids behind the plate: specialist assessment of the anchorage
bearing shear deformation, bulging, splitting, walking or seizure	Is the bearing accommodating movement and rotation as designed?	measure shear deformation and bulging with a gauge; bearing-displacement transducer: movement vs. temperature; Shore A hardness of the elastomer; level survey of the bearing seat	shear strain > 0.7, splits or exposed plates, or a movement / temperature slope well below alpha L = 0.386 mm/degC: replace the bearing (jacking under the end cross girder)

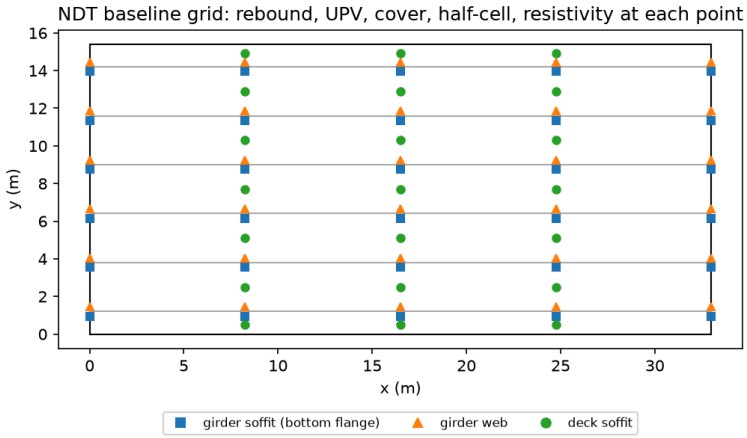


Figure 9. NDT baseline grid.

7 Deterioration scenarios and synthetic data

Six mechanisms plus the healthy state. Each has a severity s ∈ [0, 1] mapped to a physical parameter of the twin; the ranges below are my assumptions, chosen from "just detectable" to "severe but standing".

Mechanism	Model and range
prestress_loss	loss of 3-30 % of the effective force of 1 girder (30 % of cases: 2 adjacent girders); strand corrosion from grout voids / chloride. Cracks only when the soffit decompresses under frequent load (margin ~3 MPa), then EI of the mid zone drops (to 0.5 at most).
girder_cracking	flexural cracking after an overload: EI of a 9-15 m zone around mid-span of 1 girder x(1 - 0.45 s), i.e. down to 0.55 (partly cracked PSC); residual crack width 0.05-0.55 mm.
concrete_deterioration	distributed loss of concrete quality (e.g. poor compaction, ASR-type damage) in 1-3 adjacent girders: E x(1 - 0.30 s); UPV and rebound fall locally.
bearing_seized	1-6 free-end (P1) bearings restrained by debris / corrosion of the guides: horizontal stiffness 3e4-1e6 kN/m (log scale) instead of 3.3e3 (elastomer shear).
bearing_degraded	1-2 elastomeric bearings split / aged: vertical stiffness x(1 - 0.85 s).
chloride_corrosion	expansion joint at A1 or P1 leaking chloride-laden water: corrosion activity up to s in the end zones of the girders and deck below it; for s > 0.6 spalling and an end-zone EI loss of up to 20 %.

Prestress loss is applied as the reversed equivalent loads of the tendon (nodal couples ΔP·e on the girder) on the grillage: the deck shares the effect, the moment sum at mid-span equals ΔP·e (checked), and the girder only cracks under traffic when the loss consumes the ~3 MPa frequent-load margin to cracking found in Chapter 3. With HDPE ducts the strands are electrically insulated: strand corrosion from grout voids leaves no half-cell signal; only chloride-driven loss (40 % of cases) shows on the surrounding reinforcement.

A campaign is one visit of the bridge in one condition and environment (age 1–30 y, season, deck temperature, traffic). For each campaign the twin produces a 300 s, 8-channel acceleration record from which f_1, f_2, f_3 are identified by spatial modal filtering, the load-test deflections and strains of all girders, gauge-strain and camber residuals relative to the reference curves (with a creep-model error common to the bridge), bearing movement per °C, the 81 NDT readings (with per-survey calibration errors, wet-season and carbonation effects) and the inspector's observations (imperfect detection, plus unrelated defects that appear on any bridge). The dataset has **2380 campaigns** (340 per class), 192,780 NDT readings and 7,709 visual observations.

Noise-free effect of each mechanism on the measured quantities (30 °C; G1 affected; prestress loss with the chloride cause).

Mechanism	s	f_{bend}	f_{tors}	LT defl. G1 (mm)	Camber change (mm)	VW strain (με)	UPV min (km/s)	Half-cell min (mV)	Bearing mm/°C
healthy	0.00	3.442	3.510	8.37	0.00	0	4.60	-110	0.385
prestress_loss	0.25	3.442	3.510	8.37	2.44	32	4.60	-176	0.385
prestress_loss	0.50	3.442	3.510	8.37	4.12	55	4.60	-205	0.385
prestress_loss	0.75	3.442	3.510	8.37	5.81	77	4.60	-234	0.385
girder_cracking	0.25	3.405	3.482	8.77	0.00	0	4.51	-110	0.385
girder_cracking	0.50	3.342	3.473	9.24	0.00	0	4.43	-110	0.385
girder_cracking	0.75	3.266	3.470	9.80	0.00	0	4.34	-110	0.385
concrete_deterioration	0.25	3.371	3.481	8.92	0.00	0	4.25	-110	0.385
concrete_deterioration	0.50	3.275	3.474	9.55	0.00	0	3.91	-110	0.385
concrete_deterioration	0.75	3.172	3.471	10.26	0.00	0	3.56	-110	0.385
bearing_seized	0.25	3.498	3.559	8.06	0.00	0	4.60	-110	0.365
bearing_seized	0.50	3.573	3.626	7.69	0.00	0	4.60	-110	0.339
bearing_seized	0.75	3.728	3.765	7.00	0.00	0	4.60	-110	0.290
bearing_degraded	0.25	3.441	3.507	8.38	0.00	0	4.60	-110	0.385
bearing_degraded	0.50	3.439	3.504	8.39	0.00	0	4.60	-110	0.385
bearing_degraded	0.75	3.436	3.499	8.41	0.00	0	4.60	-110	0.385
chloride_corrosion	0.25	3.442	3.510	8.37	0.00	0	4.60	-205	0.385
chloride_corrosion	0.50	3.442	3.510	8.37	0.00	0	4.60	-300	0.385
chloride_corrosion	0.75	3.441	3.509	8.37	0.00	0	4.39	-395	0.385

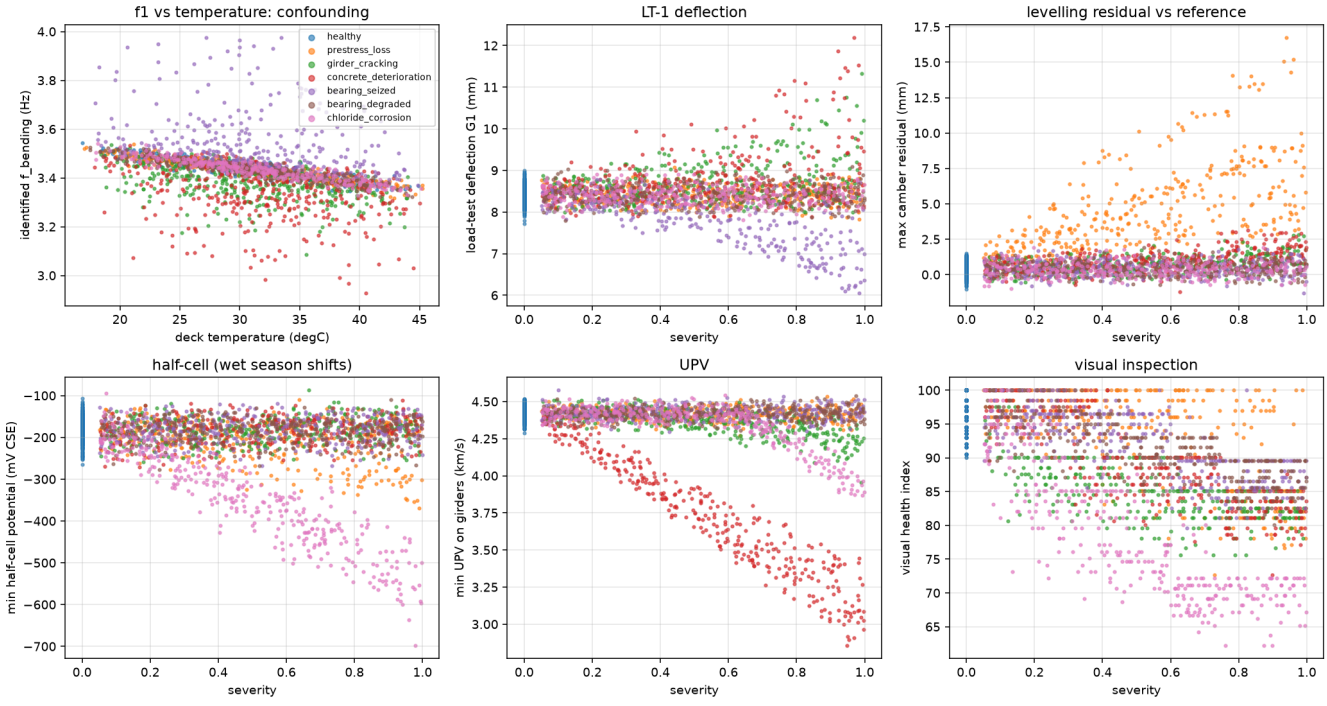


Figure 10. Synthetic campaigns: temperature confounds the frequency; each mechanism shows in a different measurement.

No single measurement sees every mechanism: prestress loss is invisible to vibration and to the load test and shows only in the levelling and gauge residuals; chloride corrosion shows only in half-cell potentials and visual inspection; a degraded bearing changes the frequency by about 0.2 % and is found essentially by the inspector.

8 Feature extraction

Features were chosen for their physical meaning, and everything that defines “normal” is learnt from **healthy training campaigns only** (in practice the first years of monitoring of this bridge). Temperature compensation of the identified frequency: f_1 = 3.4415 -0.00574(T − 30) Hz (-0.167 %/°C, R^2 = 0.81); the healthy scatter of f_1 drops from 37.8 to 16.6 mHz. Load-test deflections and strains are compensated the same way and expressed as ratios to the healthy reference of each girder; half-cell potentials and resistivity are corrected for the wet season (-39 mV), rebound for carbonation hardening. The feature vector has five groups: vibration (Δf_{1-3} in %, f_2/f_1, a_{RMS}, a_{max}), load test, long-term monitoring (camber and strain residuals, bearing movement ratio), NDT summaries and visual summaries.

Class signatures (mean features). They are physically reasonable: stiffness loss lowers f and raises the deflection, a seized bearing does the opposite, prestress loss shows only in the camber, chloride only in the potentials.

Class	Δf_1 %	Δf_2 %	LT ratio max	LT ratio min	Camber res. max (mm)	Bearing ratio	UPV min	Half-cell min (mV)	Visual HI
healthy	-0.02	-0.01	1.012	0.987	0.40	0.989	4.42	-169	97.4
prestress loss	-0.03	-0.08	1.012	0.986	4.76	0.990	4.42	-199	93.2
girder cracking	-1.87	-1.24	1.070	1.002	0.76	0.989	4.36	-170	85.3
concrete deterioration	-3.03	-1.66	1.097	1.014	0.97	0.990	3.72	-171	88.9
bearing seized	3.01	2.96	0.954	0.909	0.40	0.869	4.43	-170	92.1
bearing degraded	-0.10	-0.20	1.014	0.988	0.43	0.989	4.42	-172	91.4
chloride corrosion	-0.03	-0.04	1.011	0.985	0.38	0.989	4.34	-340	78.2

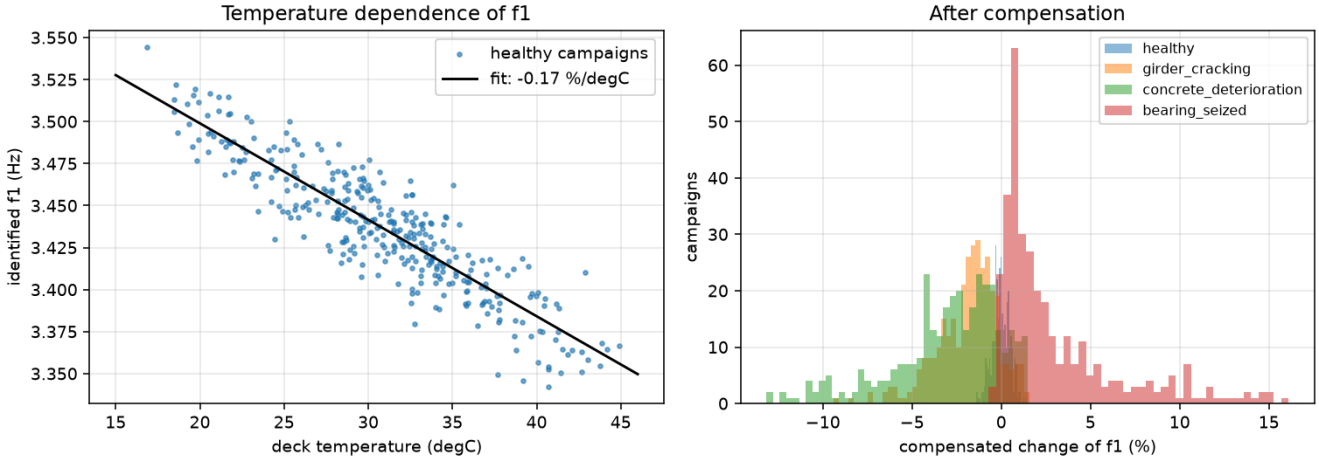


Figure 11. Temperature compensation of f_1 learnt from healthy campaigns.

9 AI/ML-assisted condition assessment

Data: 1666 training and 714 test campaigns (stratified split). Methods: anomaly detection (Mahalanobis distance with shrinkage covariance, threshold at the 99th percentile of out-of-fold healthy scores; isolation forest for comparison), classification of the mechanism (random forest), regression of severity and prestress loss (gradient boosting), localisation of the damaged girder, and k-means clustering without labels.

9.1 Is something wrong? Anomaly detection trained on the healthy bridge only

Features	False alarms %	Detected %	AUC	prestress loss	girder cracking	concrete deterioration	bearing seized	bearing degraded	chloride corrosion
vibration	1.0	36.4	0.726	3	61	72	72	11	1
load_test	1.0	29.9	0.654	1	52	70	55	2	0
monitoring	2.0	29.1	0.648	88	4	25	55	2	0
ndt	1.0	39.5	0.700	28	24	94	0	0	91
all	0.0	88.6	0.951	86	94	92	87	76	95
isolation_forest_all	0.0	47.7	0.874						

Each source alone detects only the mechanisms it can physically see; all sources together detect 89 % of damaged campaigns at 0.0 % false alarms. Features that never change on the healthy bridge (a recorded flexural crack, a bearing defect, chloride at the end blocks) have almost zero healthy scatter, so a single such observation gives Mahalanobis distances of 10^4–10^6 against a threshold of 82: the size of the score means "never seen on this bridge when healthy", not a proportionally worse condition.

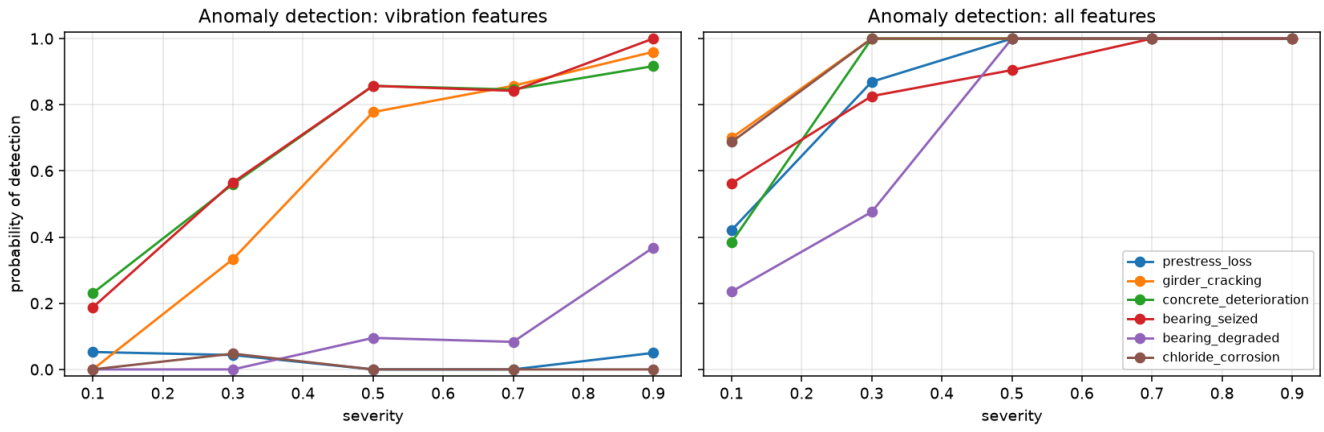


Figure 12. Probability of detection vs. severity: vibration features alone vs. all sources.

9.2 What is it? Damage type by data source

Features	Macro-F1	recall healthy	recall prestress loss	recall girder cracking	recall concrete deterioration	recall bearing seized	recall bearing degraded	recall chloride corrosion
vibration	0.409	0.37	0.20	0.45	0.53	0.85	0.25	0.23
load_test	0.379	0.22	0.23	0.49	0.56	0.73	0.20	0.25
monitoring	0.409	0.22	0.90	0.20	0.33	0.71	0.21	0.26
ndt	0.509	0.25	0.40	0.39	0.90	0.37	0.22	0.94
visual	0.751	0.94	0.35	0.92	0.71	0.68	0.64	0.96
all	0.909	0.92	0.95	0.94	0.96	0.83	0.75	0.97

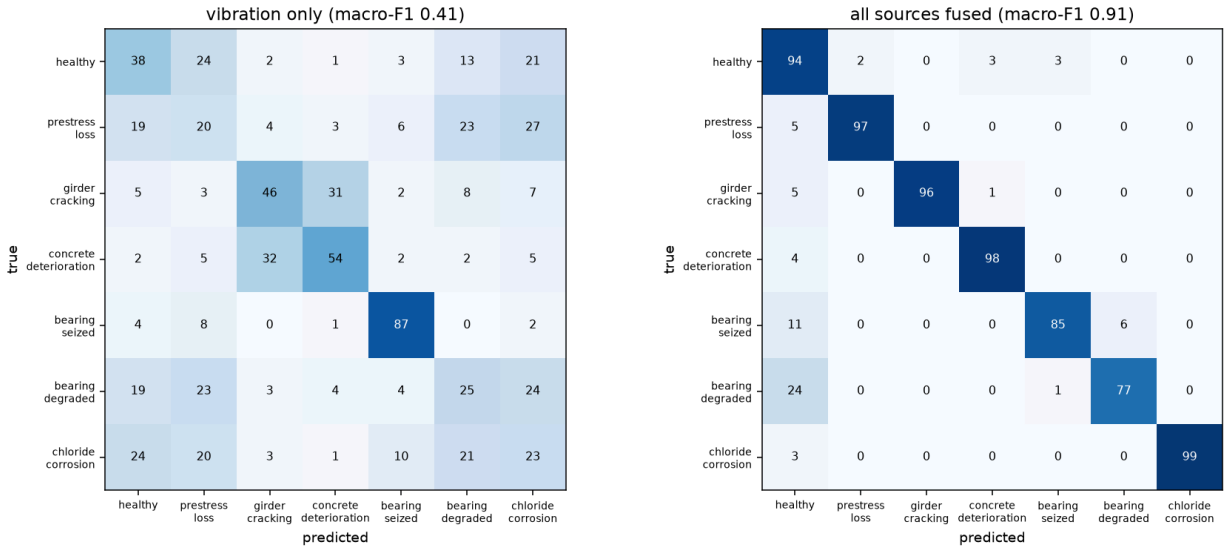


Figure 13. Confusion matrices, test set: vibration only vs. all sources fused.

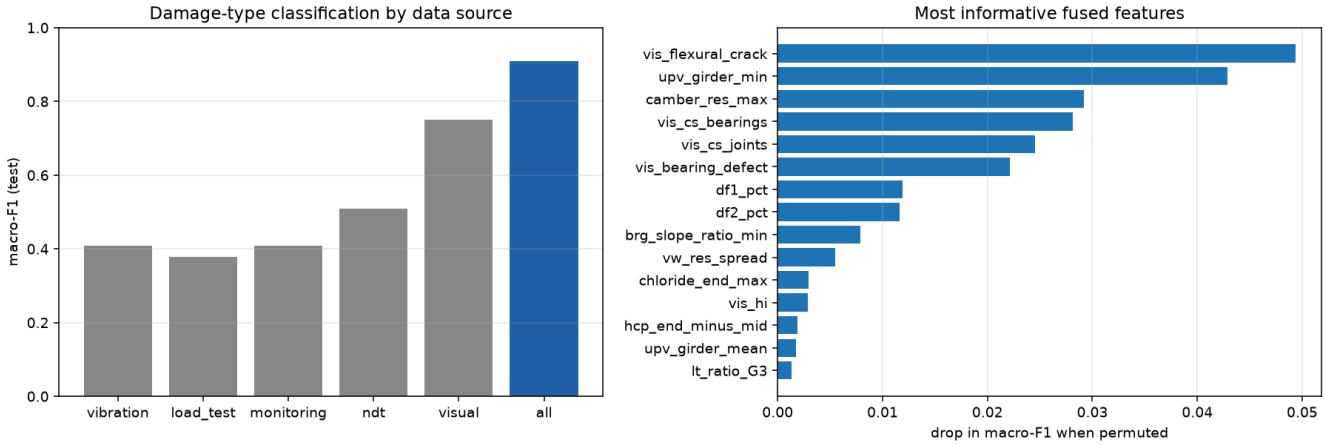


Figure 14. Classification quality by source and the most informative fused features (permutation importance).

What the fused classifier relies on.

Feature	Drop in macro-F1	Physical meaning
vis_flexural_crack	0.049	inspector recorded transverse flexural cracks: the girder has cracked under load
upv_girder_min	0.043	lowest pulse velocity on the girders: cracked or poor-quality concrete
camber_res_max	0.029	sag beyond the long-term reference curve: loss of prestress (or stiffness under permanent load)
vis_cs_bearings	0.028	worst condition state given to a bearing
vis_cs_joints	0.025	worst condition state of the expansion joints (leaking seal = source of chloride)
vis_bearing_defect	0.022	inspector recorded a bearing defect
df1_pct	0.012	change of the 1st bending frequency after temperature compensation: global flexural stiffness / boundary restraint
df2_pct	0.012	change of the 1st torsion frequency: torsional stiffness, asymmetric damage, bearings
brg_slope_ratio_min	0.008	bearing movement per degC / alpha L: falls when a free bearing is restrained
vw_res_spread	0.005	difference between girders of the embedded-gauge strain residual: prestress lost in one girder only

9.3 How bad, and where?

Regression on test campaigns of each class (prestress loss in % of P_e).

Target	MAE	R^2
prestress loss from monitoring	3.599	0.62
prestress loss from vibration	7.420	-0.37
prestress loss from all	3.205	0.70
severity prestress loss	0.119	0.70
severity girder cracking	0.084	0.88
severity concrete deterioration	0.041	0.96
severity bearing seized	0.069	0.89
severity bearing degraded	0.079	0.88
severity chloride corrosion	0.017	0.99

Localisation of the damaged girder (chance: 17 %).

Mechanism	Exact girder %	Within one girder %
prestress_loss	75	99
girder_cracking	61	82
concrete_deterioration	61	89

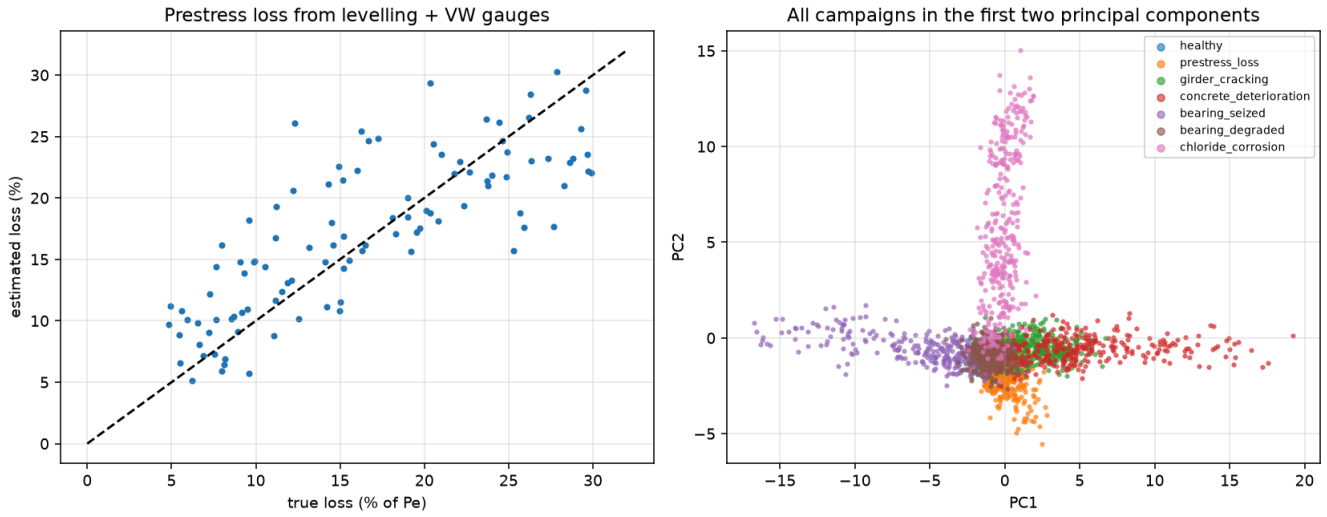


Figure 15. Prestress loss estimated from levelling + VW gauges, and all campaigns in the first two principal components.

Clustering. k-means with seven clusters on all features reaches an adjusted Rand index of 0.25 with the true classes: mechanisms with a strong signature form their own clusters, while degraded bearings and early-stage damage merge with the healthy campaigns – without labels the data separate “clearly changed” from “normal”, not mechanisms.

9.4 What the results mean for the engineer

- Vibration alone is a weak assessor (macro-F1 0.41): it sees stiffness changes and seized bearings, but not prestress loss or corrosion, and temperature moves f_1 as much as moderate damage. Fusing all sources raises macro-F1 to **0.91**.
- Each source answers a different question: visual inspection finds bearings, joints and surface defects; NDT finds corrosion and poor concrete; levelling and embedded gauges are the only route to prestress loss; vibration and the load test measure stiffness and restraint. This is exactly the fusion asked for in the maintenance chapter.

- The outputs are physically reasonable (class signatures above), but they are **digital-twin performance**: the deterioration ranges, noise levels and detection probabilities are my assumptions. On a real bridge the healthy reference must be learnt from its own first years of monitoring, and the classifier needs field cases before it can be trusted for anything but triage.
- The ML output is a recommendation for the engineer (which mechanism is most likely, how severe, where to look), not a decision. Chapter 10 turns it into decisions with explicit rules that the engineer can audit.

10 Maintenance scheme

The decision fuses **visual inspection + NDT + structural response + vibration + AI/ML**. Rules that need no model come first (visual CS4 on girders or bearings; CS3 on structural components or joints; half-cell below −350 mV; UPV below 3.75 km/s; camber more than 3 mm below the reference curve; bearing movement below 80 % of αL per °C), then the ML assessment (anomaly flag; most likely mechanism with probability ≥ 0.6 and estimated severity ≥ 0.4 → repair; structural mechanism with severity ≥ 0.8 or estimated prestress loss > 20 % → urgent). Every decision lists its reasons.

Level	Decision	Re-inspection
0	routine monitoring	routine cycle
1	enhanced monitoring	targeted NDT within 3 months; monitoring review monthly
2	detailed investigation	investigation within 1 month
3	maintenance / repair	repair within 6 months; re-inspect after repair and at 1 year
4	urgent: restrict loading	immediately

Mechanism	Detailed investigation	Maintenance / repair
prestress_loss	drill-and-endoscope the ducts of the suspected girder at L/4, L/2, 3L/4; GPR / ultrasonic tomography for grout voids; load test with crack-opening gauges (decompression load)	vacuum re-grouting of voids; if the estimated loss exceeds ~10 % of Pe: external post-tensioning (deviators at the cross girders) designed for the lost force; reinstate monitoring targets
girder_cracking	crack mapping and crack gauges, UPV across cracks, load test; identify the overload (weigh-in-motion records) and check ULS with the reduced stiffness	epoxy injection of cracks wider than 0.2 mm; bonded CFRP on the soffit if the capacity check with the cracked section fails; enforce load limits
concrete_deterioration	cores (IS 516 Part 4) to calibrate rebound + UPV, petrography, extent mapping	remove and patch deteriorated concrete, surface protection; strength assessment with in-situ strength before deciding on strengthening
bearing_seized	close visual of the free bearings, measure movement vs. temperature over a day	clean debris, free or replace the bearing (jacks under the end cross girder, 4 per girder end)
bearing_degraded	measure bulging, splits and shear strain; Shore A hardness	replace the elastomeric bearing (jacking under the end cross girder at the design jacking points)
chloride_corrosion	chloride profiles and half-cell grid on the end blocks under the joint; open one anchorage pocket	replace the expansion-joint seal first (stop the source); patch repair of spalls; silane on end blocks; galvanic / impressed-current cathodic protection if potentials stay below -350 mV

Is the logic safe and economical? On the 19 condition groups of the test set: 100 % of healthy campaigns stay on routine or enhanced monitoring and 0.0 % are sent to repair; 0.0 % of severe cases (s ≥ 0.6) are left on routine monitoring and 100 % are investigated or repaired.

Share of test campaigns at each decision level.

Condition	Severity	L0	L1	L2	L3	L4
bearing degraded	low	0.76	0.00	0.24	0.00	0.00
bearing degraded	medium	0.15	0.00	0.38	0.47	0.00
bearing degraded	high	0.00	0.00	0.00	0.37	0.63
bearing seized	low	0.37	0.15	0.48	0.00	0.00
bearing seized	medium	0.09	0.03	0.24	0.64	0.00
bearing seized	high	0.00	0.00	0.02	0.00	0.98
chloride corrosion	low	0.10	0.00	0.90	0.00	0.00
chloride corrosion	medium	0.00	0.00	0.24	0.76	0.00
chloride corrosion	high	0.00	0.00	0.00	0.00	1.00
concrete deterioration	low	0.35	0.00	0.65	0.00	0.00
concrete deterioration	medium	0.00	0.00	0.48	0.52	0.00
concrete deterioration	high	0.00	0.00	0.00	0.32	0.68
girder cracking	low	0.22	0.00	0.74	0.04	0.00
girder cracking	medium	0.00	0.00	0.48	0.24	0.28
girder cracking	high	0.00	0.00	0.00	0.00	1.00
healthy	healthy	1.00	0.00	0.00	0.00	0.00
prestress loss	low	0.41	0.03	0.44	0.12	0.00
prestress loss	medium	0.03	0.00	0.22	0.44	0.31
prestress loss	high	0.00	0.00	0.00	0.21	0.79

Example decision (most severe prestress-loss test campaign): *urgent: restrict loading* — reasons: visual CS4 on girders / bearings; visual CS3 on a structural component or joint; camber 5.8 mm below the reference curve; anomaly score 446182 > 82; ML: prestress_loss (p = 1.00), severity ~0.87; estimated prestress loss 22 %; action: restrict loading; drill-and-endoscope the ducts of the suspected girder at L/4, L/2, 3L/4; GPR / ultrasonic tomography for grout voids; load test with crack-opening gauges (decompression load); re-inspection: immediately.

10.1 Thirty years of this bridge

The same twin was run year by year: the A1 joint starts leaking in year 8 (chloride corrosion of the end zones grows by 0.09 per year), an overload cracks girder G2 in year 18, and prestress loss from grout voids starts in G4 in year 23. A **proactive owner** (yearly monitoring + NDT + ML, repairs at level 3, or level 2 confirmed in two consecutive years) is compared with a **reactive owner** (visual inspection only; acts on CS4 or a visual health index below 60). Costs are relative units I assumed (repair cost growing with the square of the severity, monitoring 0.3 per year, investigation 0.5, load restriction 5 per year while the severity exceeds 0.8). Damage still unrepaired at year 30 is charged at its repair cost: it is deferred expenditure, not a saving.

Owner	Repairs (year, true severity)	Worst severity	Spent	Unrepaired at year 30	Total
proactive	chloride corrosion (year 10, s = 0.27); girder cracking (year 18, s = 0.45); prestress loss (year 27, s = 0.60)	0.60	31.7	0.0	**31.7**
reactive	chloride corrosion (year 14, s = 0.63); girder cracking (year 20, s = 0.45)	0.96	20.8	30.6	**51.4**

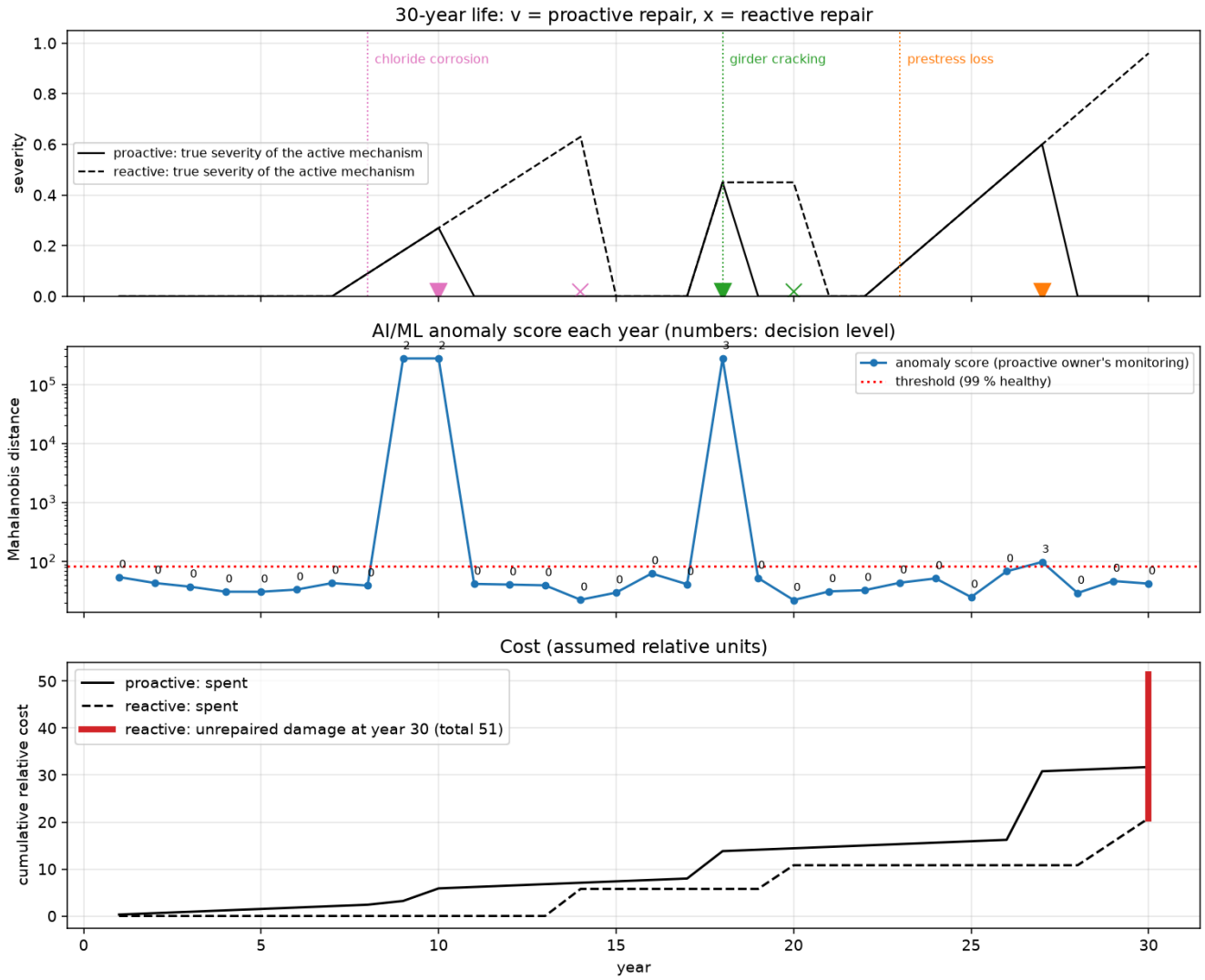


Figure 16. Thirty-year life: true severity, repairs, the yearly anomaly score with the decision level, and cumulative cost.

- **Chloride corrosion** is caught by the half-cell survey and the ML one year after it starts and repaired at a low severity; the reactive owner waits until the visual condition is poor, when the patch area and the repair cost are much larger.
- **The overload crack** in G2 is visible to both owners; the difference is only how quickly they act.
- **Prestress loss from grout voids** is the decisive case. It is invisible to the inspector (lowest visual health index during it: 92), so the reactive owner never repairs it and ends the period at severity 0.96 (about 29 % loss in G4) under a load restriction. The proactive owner's levelling and embedded gauges flag it in year 27 (severity 0.60). This is later than one would like: in the first years the classifier already leaned towards prestress loss but the anomaly score stayed below the 99 % threshold. Gradual loss of a few per cent is within the scatter of the long-term reference curve, which is the practical detection limit of this scheme.
- Over the 30 years the proactive owner's total is 32 against 51 relative units, and the bridge never exceeds severity 0.60. The comparison depends on the assumed costs; the robust conclusion is the safety one: without monitoring, the prestress loss is found only when it is already severe.

11 Use of AI and verification

An AI assistant (Claude, Anthropic) was used, as permitted by §1.15, to formulate the problem, write the Python tools, generate the synthetic data and draft this report. The engineering choices (span, configuration, two-stage stressing, strand count, damage ranges) were discussed and decided step by step, and every computational result was checked independently:

- **29 of 29 automated checks pass** (`tests/`): single-beam results against PL/4, PL³/48EI and wL²/8; grillage equilibrium; vehicle search; friction and wedge-slip against the closed form; ULS against the parabola–rectangle hand formula; zero eigen-stress for uniform and linear temperature; mass of the twin against the dead loads; f_1 against the beam formula; FDD on a known SDOF signal; prestress-loss equivalent loads against ΔP·e; JS and Python inspection rules on 500+ cases; detector false-alarm rate on fresh data.
- Hand checks reported in the text: Courbon distribution, simply supported beam frequency, dead-load mass.
- Errors found during the work and corrected: a timeline constant that silently overwrote the deck thickness (a test now guards the configuration); a first prestress design (3 × 12T15) that cracked under the frequent combination; a duct clash of 183 mm against 190 mm required; an optimistic NDT model in which half-cell potentials “saw” strands inside insulating HDPE ducts, and an anomaly threshold set on in-sample scores that gave 8 % instead of ~1 % false alarms.

> [Student: add your own account here — which AI suggestions you questioned or changed, what you checked by hand, and what you learnt. This section must be in your own words.]

12 Assumptions and limitations

- Span S1 of a viaduct; piers, abutments, foundations, seismic and wind design outside the scope.
- IRC:6-2017 loads; the 10 % multi-lane reduction only for three Class A lanes; non-urban footpath intensity.
- Loss parameters: $\mu = 0.17$, $k = 0.002$/m, wedge slip 6 mm, RH 70 %, final relaxation 3 × the 1000-h value; creep and shrinkage by IRC:112 Annex A2 (drying shrinkage by the EN 1992-1-1 Annex B model with $f_{ck,cyl} = 0.8\ f_{ck,cube}$).
- Characteristic prestress 0.9/1.1 P_m; SLS combination factors: live load 0.75 frequent, temperature 0.6 / 0.5 / 0.5.
- Cable radii, anchorage spacing and edge distances to be confirmed against the approved PT system.
- Twin: $E_d = (E_c + 19)/1.25$, −0.35 %/°C change of E with temperature, 1.5 % damping, barriers non-composite.
- Deterioration ranges, NDT and inspection error models, traffic and ambient levels, and costs are assumptions; all data are synthetic, so the ML results are digital-twin performance and need field recalibration.
- Code values quoted from memory (e.g. IRC:112 Table 7.1, IS 516 UPV grading) are to be verified against the current editions.

References

1. IRC:5-2015 Standard specifications and code of practice for road bridges, Section I.
2. IRC:6-2017 Section II: Loads and load combinations.
3. IRC:112-2020 Code of practice for concrete road bridges.
4. IRC:83 (Part II)-2018 Elastomeric bearings.
5. IRC:SP:35 Guidelines for inspection and maintenance of bridges; IRC:SP:51 Guidelines for load testing of bridges.
6. IS 516 (Part 4, Part 5 Sec 1, Sec 4) Hardened concrete: cores, ultrasonic pulse velocity, rebound hammer.
7. ASTM C876 Corrosion potentials of uncoated reinforcing steel in concrete.
8. BS 8110-2:1985 cl. 7.2 (dynamic and static modulus); EN 1992-1-1 Annex B (creep and shrinkage).
9. Hambly, E.C. Bridge Deck Behaviour, 2nd ed. (grillage analysis).
10. Brincker, R., Zhang, L., Andersen, P. (2001) Modal identification of output-only systems using frequency domain decomposition. Smart Materials and Structures 10, 441–445.

Code and data: `bridge/` (modules), `run_step1_analysis.py` … `run_step7_maintenance.py`, `outputs/` (JSON results, figures, datasets), `inspection_app/`, `tests/`, `report/build_report.py`.